1. Problem Statement

This project focuses on analyzing historical temperature data across a country, spanning over a century, to identify patterns and trends in temperature changes. The dataset includes time-stamped records of average temperatures, associated uncertainties

2. Dataset

Source: Kaggle(https://www.kaggle.com/datasets/berkeleyearth/climate-change-earth-surface-temperature-data/code)

The Dataset contains Global Land and Ocean Temperatures with data from all major countries. We selected the Global Temperatures for our project. Upon review with our professor, we condensed the dataset and started from the year 1900 as the there was too much data and used data for one location.

Features:

- Historical Temperature records: Average temperatures recorded over time for various cities within the country.
- Uncertainty Values: Accompanying uncertainty estimates for each temperature measurement.
- Geographic Metadata: Includes city names, latitude, and longitude for spatial analysis.
- Time Series Structure: Data organized by dates, enabling temporal analysis of trends and forecasting.

The following dataset has been used or a variety of purposes related to analyzing and understanding temperature trends globally and regionally. Researchers and analysts have used it for exploratory data analysis to uncover temperature trends, seasonality, and anomalies across global and regional scales. The dataset has also been applied to assess global warming by comparing historical temperature data to contemporary trends, often visualized in maps or through statistical measures.

3. Methodology

This analysis focuses on temperature trends within a specific country, using the **ARIMA**, **SARIMAX**, and **ETS models** to forecast future temperature changes. The objective is to capture the unique climate dynamics of the country, assess the impact of climate change, and provide reliable long-term temperature forecasts. ARIMA models linear trends, SARIMAX incorporates seasonal patterns and external factors, and ETS is used for capturing non-linear seasonal variations. Together, these models offer insights into future temperature trajectories and help in climate policy planning.

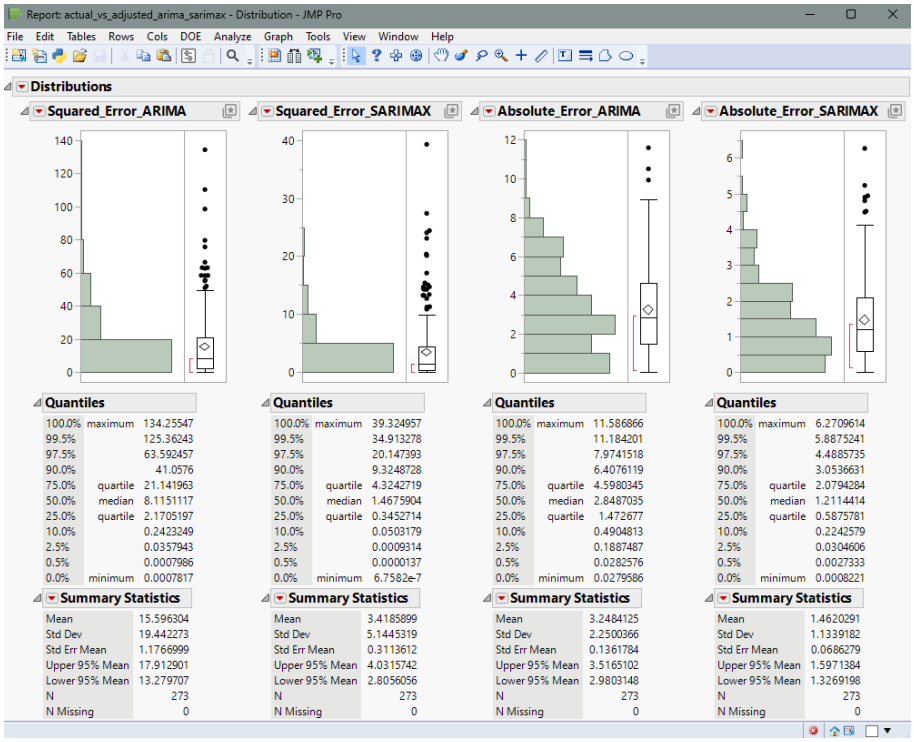
3.1 Model 1 – ARIMA & SARIMAX

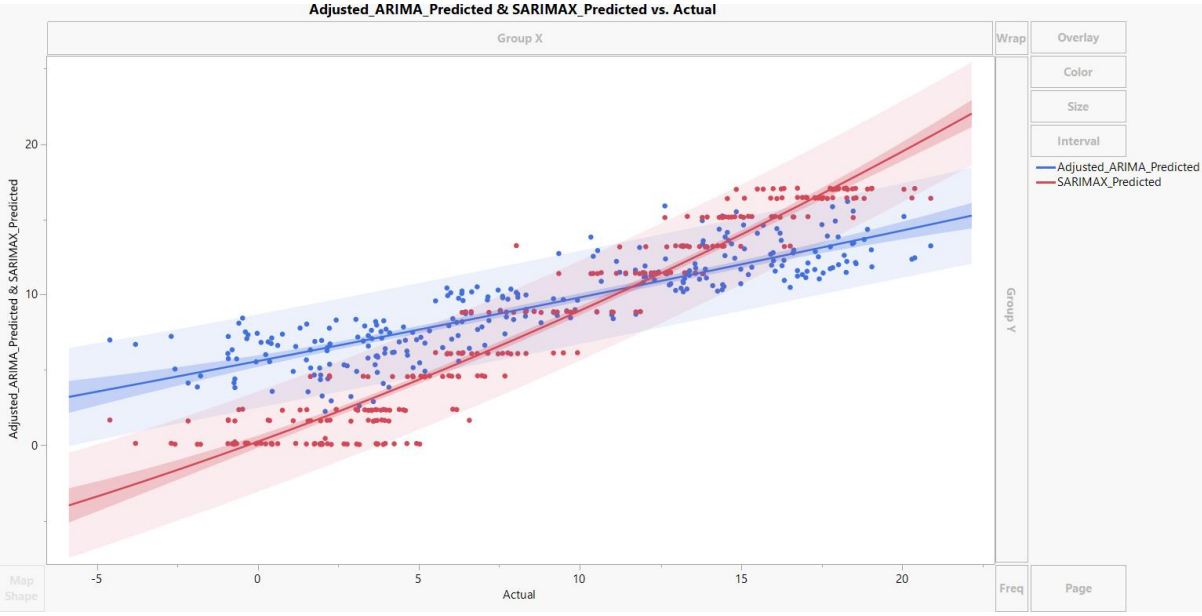
We Performed time series forecasting on temperature data using ARIMA and SARIMAX models. Assessed the models' performance and adjust them as needed to improve forecast accuracy, considering the seasonal nature of the dataset.

The dataset used in this analysis contains monthly temperature data (AverageTemperature) from 1900 onwards, which was split into two subsets: 80% of the data was allocated to the training set, while the remaining 20% was used for testing. The process began with **initial setup and preprocessing**, where the dataset was loaded, dates were parsed and set as the index, and the AverageTemperature column was selected for modeling. The data was then split into training and testing sets.
In the **initial ARIMA modeling** phase, an ARIMA model with parameters (p=1, d=1, q=1) was fitted to the data.

However, the model's performance was suboptimal due to the presence of strong seasonality in the dataset, which ARIMA does not explicitly handle, resulting in inaccurate forecasts. To improve this, the ARIMA model parameters were adjusted to (p=2, d=1, q=2) in an effort to better capture the underlying patterns in the data. While this adjustment led to some improvement, it still struggled with the pronounced seasonal patterns present in the dataset.

To address these challenges, the SARIMAX model was introduced. SARIMAX explicitly models the seasonal components by incorporating seasonal parameters (P=1, D=1, Q=1, s=12) to account for monthly seasonality. This model significantly outperformed the adjusted ARIMA model, providing more accurate forecasts and capturing the seasonal behavior of temperature trends effectively.





The following comparisons show actual vs. predicted values for both models

The diagonal comparison of the actual vs. SARIMAX forecasts, when contrasted with the actual vs. improved ARIMA model, indicates that the SARIMAX model performs better at predicting the temperature trend. This suggests that SARIMAX is more effective in capturing the underlying patterns and seasonality in the data.

1. **Initial ARIMA**: The initial ARIMA model struggled to handle the seasonality in the data, leading to poor performance and inaccurate forecasts.
2. **Adjusted ARIMA**: Tuning the ARIMA parameters improved the model's performance, but it still failed to adequately capture the seasonal patterns.
3. **SARIMAX**: By explicitly modeling seasonality, SARIMAX significantly reduced forecasting errors and provided a much better fit and more accurate predictions.

3.2 Model 2 – ETS

The same dataset was used which was split into training and testing sets—80% of the data for training and 20% for testing.
1. Initial Setup and Preprocessing: The dataset was loaded, with dates parsed and set as the index for time-based analysis. The focus was on the AverageTemperature column, which was used for forecasting. The data was then split into the training and testing sets.

2. ETS Model Implementation: The Exponential Smoothing State Space Model (ETS) was implemented, which automatically selects the optimal combination of error, trend, and seasonal components for the dataset. ETS is ideal for time series data with trends and seasonal variations, making it well-suited for temperature forecasting. The model was trained using the training data and then applied to predict future temperatures during the testing period.

The ETS model effectively captured both the trend and seasonality in the temperature data. The model's performance metrics indicated a reasonably good fit, with low errors across all evaluation metrics. This suggests that the ETS model was successful in reflecting the underlying patterns in the historical temperature data. The forecasts produced by the model closely aligned with actual temperature values, demonstrating its ability to accurately predict future trends while maintaining a strong alignment with observed data.

Model: Seasonal Exponential Smoothing(12, Zero to One)

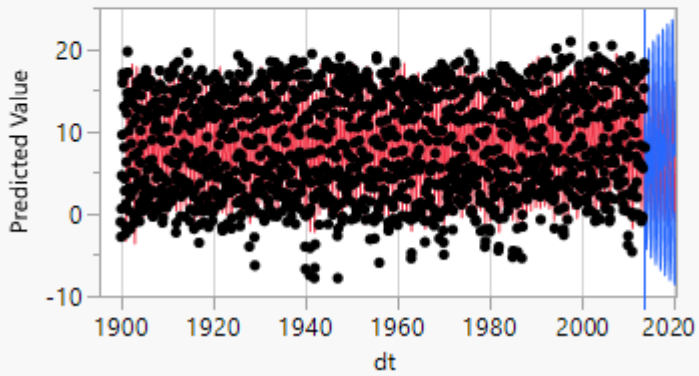
Model Summary

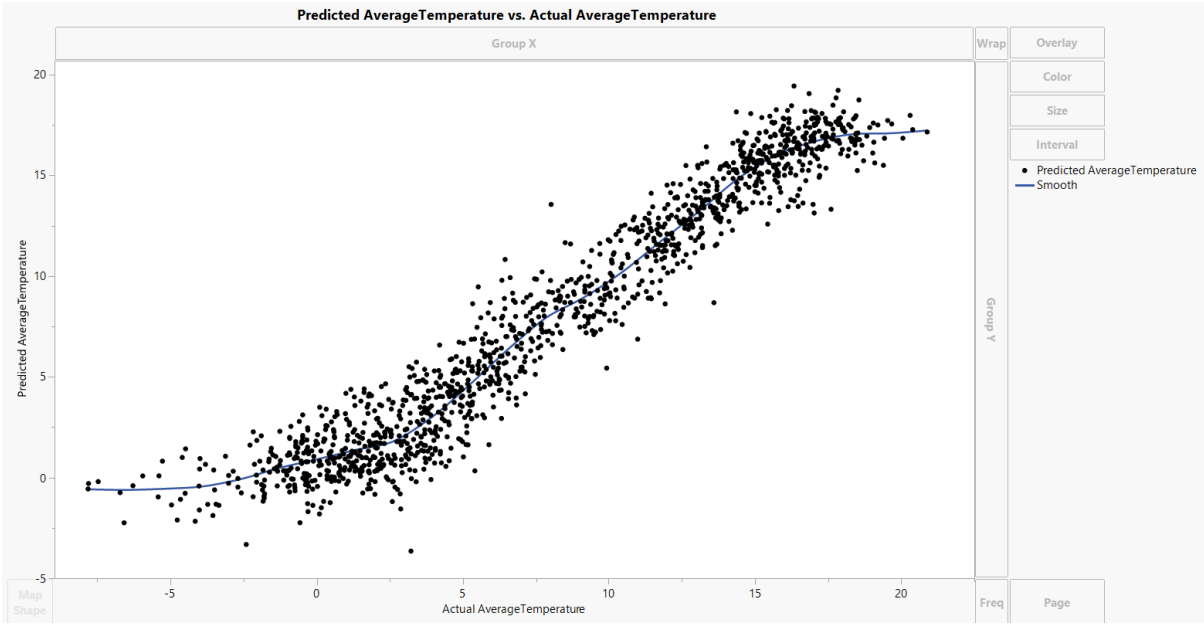
DF	1350	Stable	Yes
Sum of Squared Innovations	3584.12808	Invertible	No
Sum of Squared Residuals	3730.82018		
Variance Estimate	2.65490969		
Standard Deviation	1.62938936		
Akaike's 'A' Information Criterion	5216.57441		
Schwarz's Bayesian Criterion	5226.99309		
RSquare	0.93199041		
RSquare Adj	0.93194003		
MAPE	75.5403117		
MAE	1.27345246		
-2LogLikelihood	5212.57441		

Parameter Estimates

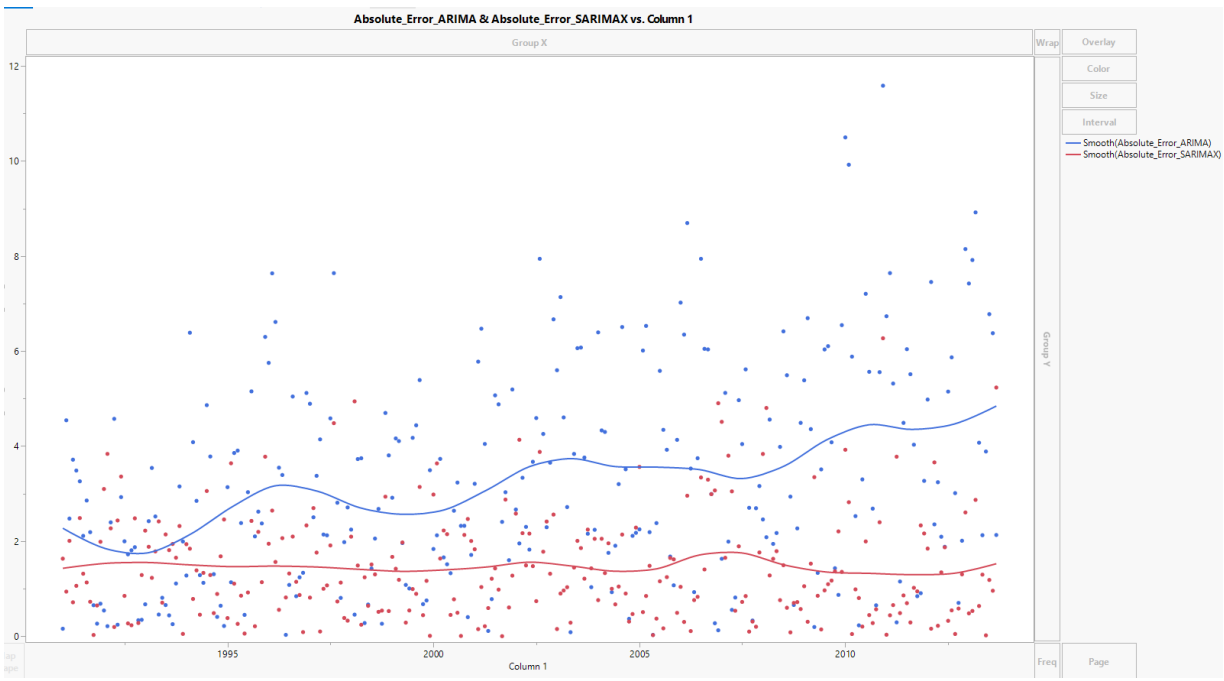
Term	Estimate	Std Error	t Ratio	Prob>|t|
Level Smoothing Weight	0.25686186	0.0420626	6.11	<.0001*
Seasonal Smoothing Weight	0.00000080	0.0000012	0.70	0.4869

Forecast

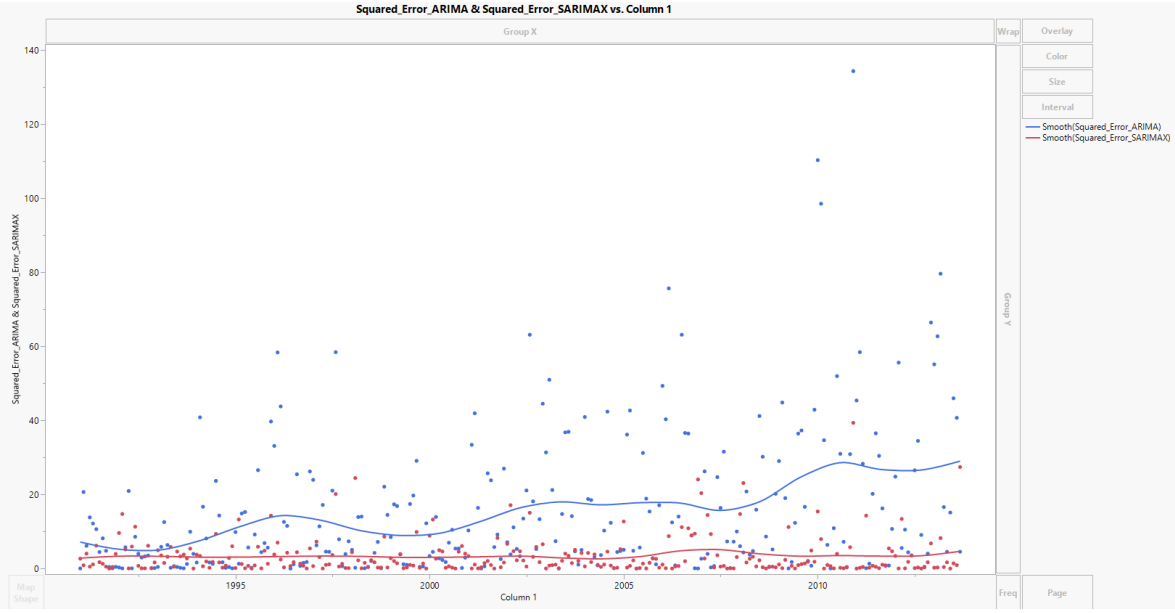




With low error metrics like MAE, MSE, RMSE, and MAPE, the ETS model performed well and successfully captured the seasonality and underlying trends in the temperature data. The ETS model is a good option for forecasting in situations where time series have both trend and seasonal components because of its capacity to handle such data. To guarantee the best forecasting accuracy, there is still opportunity for improvement, such as adjusting the model's parameters for a better fit or evaluating its results against those of other models, such as ARIMA or SARIMAX.



(Absolute Error- ARIMA vs SARIMAX)



(Squared Error – ARIMA vs SARIMAX)

Combined Results:

Metric	Initial ARIMA(1,1,1)	Adjusted ARIMA(2,1,2)	SARIMAX	ETS
MSE	Very High (Failed)	15.60	3.42	4.94
RMSE	Very High (Failed)	3.95	1.85	2.23
MAE	Very High (Failed)	3.25	1.46	2.12

4. Conclusion

This analysis produced a number of important findings and insights. Strong seasonality in the temperature data initially caused the ARIMA model to falter, requiring modifications before the SARIMAX model was finally adopted. In every metric, SARIMAX performed better than the original and modified ARIMA models, proving that seasonal data necessitates explicit seasonality modelling. For predicting time series data with seasonal patterns, SARIMAX outperformed other models due to its robust performance. A good option for time series with trend and seasonal components, the ETS model also performed well, showing low error metrics like MAE, MSE, RMSE, and MAPE. To ensure the best forecasting performance, future developments for both models might include additional parameter tweaks or comparisons with other models, like ARIMA or SARIMAX. It is advised to give models like SARIMAX or ETS precedence over

ARIMA for datasets with significant seasonal patterns in order to better capture the underlying dynamics.